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Target (series) 70^+

SENIOR INTERMEDIATE

IMP QUESTIONS LIST

Mathematics

Paper 2A (EM) (AP/TS)

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I.P.E MODEL PAPER BLUE PRINT AS PER BIE AP/TS

SECTION - A (Very Short Answer type Questions) $10 \times 2 = 20M$

1. COMPLEX NUMBERS
2. COMPLEX NUMBERS
3. DE MOIVRES THEOREM
4. QUADRATIC EXPRESSIONS
5. THEORY OF EQUATIONS
6. PERMUTATIONS
7. COMBINATIONS
8. BINOMIAL THEOREM
9. MEASURES OF DISPERSION
10. RANDOM VARIABLES & PROBABILITY DISTRIBUTION

SECTION - B (Short Answer type Questions) $5 \times 4 = 20M$

11. COMPLEX NUMBERS
12. QUADRATIC EXPRESSIONS
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15. PARTIAL FRACTIONS
16. PROBABILITY
17. PROBABILITY

SECTIONS - C (Long Answer type Questions) $5 \times 7 = 35M$

18. DE MOIVRES THEOREM
19. THEORY OF EQUATIONS
20. BINOMIAL THEOREM
21. BINOMIAL THEOREM
22. MEASURES OF DISPERSION
23. PROBABILITY
24. RANDOM VARIABLES & PROBABILITY DISTRIBUTION

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QNO : 1 & 2 IN IPE (AP/TS)

1. COMPLEX NUMBERS

1. Express the following in the form of a+ib
   i). $3(7+7i)+i(7+7i)$ ii). $(2-3i)(3+4i)$ iii). $(1+2i)^3$ iv). $(1-i)^3(1+i)$
   v). $\frac{2+5i}{3-2i}+\frac{2-5i}{3+2i}$ vi). $\frac{a-ib}{a+ib}$ vii). $\frac{4+3i}{(2+3i)(4-3i)}$
2. Find the additive inverse of
   i). $(\sqrt{3},5)$ ii). $(-6,5)+(10,-4)$ iii). $(2,1)(-4,6)$
3. Find the multiplicative inverse of
   i). $(3,4)$ ii). $7+24i$ iii). $\sqrt{5}+3i$ iv). $(\sin\theta, \cos\theta)$
4. Find the complex conjugate of
   i). $3+4i$ ii). $(15+3i)-(4-20i)$ iii). $(3+4i)(2-3i)$ iv). $(2+5i)(-4+6i)$
   v). $\frac{5i}{7+i}$
5. Find the real and imaginary parts of the complex number $\frac{a+ib}{a-ib}$
6. If $z = (\cos\theta, \sin\theta)$, then find the value of $z - \frac{1}{z}$
7. Find the value of $i^2+i^4+i^6+$.......(2n+1)terms
8. Show that the least positive integral value of 'n' for which $\left(\frac{1+i}{1-i}\right)^n = 1$ is 4
9. If $z = 2-3i$, show that $z^2-4z+13 = 0$
10. Find the square root of the following
    i). $3+4i$ ii). $-5+12i$ iii). $7+24i$ iv). $-8 - 6i$
11. i). If $(a+ib)^2 = x + iy$, then find x^2+y^2
    ii). If $(\sqrt{3}+i)^{100} = 2^{99}(a+ib)$, show that
    iii). If $(1-i)(1-2i)(1-3i).....(1-ni) = x - iy$, prove that $2.5.10.....(1+n^2) = x^2+y^2$
    iv). If $x+iy = \text{cis}\,\alpha\ \text{cis}\,\beta$, find the value of x^2+y^2
12. Represent the complex number 2+3i in argand plane
13. i). If $\frac{\pi}{5}$, and $\frac{\pi}{3}$ are the arguments of $\overline{z_1}$, z_2 find the value of $\text{Arg}(z_1) + \text{Arg}(z_2)$
    ii). If $z_1 = -1$, and $z_2 = i$ find the value of $\arg(z_1/z_2)$
    iii). If $z_1 = -1$, $z_2 = -i$, find $\text{Arg}(z_1.z_2)$
    iv). If $z = \frac{1+2i}{1-(1-i)^2}$ then find Argz
14. Express the following in the mod - amplitude form (polar form)
    i). $1+\sqrt{3}i$ ii). $-\sqrt{3}+i$ iii). $-1-\sqrt{3}i$ iv). $1- i$
    v). $-\sqrt{7}+i\sqrt{21}$ vi). $-1 -i$

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15. If $\sqrt{3}+i = r(\cos\theta + i\sin\theta)$ find the value of θ
16. If $(\cos 2\alpha + i\sin 2\alpha)(\cos 2\beta + i\sin 2\beta) = \cos\theta + i\sin\theta$.then find the value of θ.
17. Find the locus of $z = x+iy$,

i). if $\operatorname{Arg} z = \frac{\pi}{4}$ ii). if $\arg(z-1) = \frac{\pi}{2}$ iii). if $|z| = 1$ iv). if $|z| = 2$

v). if $\operatorname{Img}(z^2) = 4$

QNO : 3 IN IPE (AP/TS)
2. DE MOIVRES THEOREM

1. If A, B, C are the angles of a triangle and $x = \operatorname{cis}A$, $y = \operatorname{cis}B$, $z = \operatorname{cis}C$ find the value of xyz.
2. If $x = \cos\theta + i\sin\theta$, find $x^6 + \frac{1}{x^6}$
3. If $1, \omega, \omega^2$ are the cube roots of unity, show that
   i). $(1+\omega)^3+(1+\omega^2)^3 = -2$ ii). $(1-\omega+\omega^2)^5+(1+\omega-\omega^2)^5 = 32$
4. If $1, \omega, \omega^2$ are the cube roots of unity, show that
   i). $(1-\omega+\omega^2)^6 + (1+\omega-\omega^2)^6 = 128 = (1-\omega+\omega^2)^7 + (1+\omega-\omega^2)^7$
   ii). $(2-\omega)(2-\omega^2)(2-\omega^{10})(2-\omega^{11}) = 49$ iii). $(1-\omega)(1-\omega^2)(1-\omega^4)(1-\omega^8) = 9$
   iv). $(a+b)(a\omega+b\omega^2)(a\omega^2+b\omega) = a^3+b^3$
   v). $(x+y+z)(x+y\omega+z\omega^2)(x+y\omega^2+z\omega) = x^3+y^3+z^3-3xyz$
   vi). $\frac{1}{1+2\omega} - \frac{1}{1+\omega} + \frac{1}{2+\omega} = 0$
5. If $1, \omega, \omega^2$ are the cube roots of unity, then find the value of
   i). $\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2} + \frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2}$
   ii). $(a+2b)^2(a\omega+2b\omega^2)^2(a\omega^2+2b\omega)^2$ iii). $(1-\omega+\omega^2)^3$
6. Find the cube roots of 8
7. If $1, \omega, \omega^2$ are the cube roots of unity, write down the roots of $(x-1)^3+8 = 0$
8. Simply $\frac{(\cos\alpha + i\sin\alpha)^4}{(\sin\beta + i\cos\beta)^8}$
9. If $1, \omega, \omega^2$ are the cube roots of unity, and $x = \omega-\omega^2-2$ then show that $x^2+4x+7 = 0$
10. If α, β are the roots of the equation $x^2+x+1 = 0$ then prove that $\alpha^4+\beta^4+\alpha^{-1}\beta^{-1} = 0$
11. Find the value of
    i). $(1+\sqrt{3}\,i)^3$ ii). $(1-i)^8$ iii). $(1+i)^{16}$
12. Show that $\left(\frac{\sqrt{3}}{2}+\frac{i}{2}\right)^5 - \left(\frac{\sqrt{3}}{2}-\frac{i}{2}\right)^5 = i$

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QNO : 4 IN IPE (AP/TS)
3. QUADRATIC EXPRESSIONS

1. If α, β are the roots of $ax^2+bx+c = 0$ then find the values of the following

   i). $\frac{1}{\alpha}+\frac{1}{\beta}$ ii). $\frac{1}{\alpha^2}+\frac{1}{\beta^2}$ iii). $\frac{\alpha^2+\beta^2}{\alpha^{-2}+\beta^{-2}}$ iv). $\alpha^2+\beta^2$

   v). $\alpha^3+\beta^3$ vi). $\alpha^4\beta^7+\alpha^7\beta^4$
2. Find the quadratic equation whose roots are

   i). 2, 5 ii). $2\pm\sqrt{3}$ iii). $7\pm2\sqrt{5}$ iv). $-3\pm5i$

   v). $-5\pm2\sqrt{3}$ vi). $-a\pm ib$ vii). $\frac{p-q}{p+q}, -\frac{p+q}{p-q}$ viii). $\frac{m}{n}, -\frac{n}{m}$
3. Determine the nature of the roots without finding the roots

   i). $4x^2-20x+25 = 0$ ii). $2x^2-7x+10 = 0$ iii). $2x^2-8x+3 = 0$ iv). $3x^2+7x+2 = 0$
4. i). If the equation $x^2-15- m(2x-8) = 0$ has equal roots, then find m

   ii). If the equation $x^2+2(k+2)x + 9k = 0$ has equal roots, then find the value of k.

   iii). If $x^2-2(1+3m)x+7(3+2m) = 0$ has equal roots, then find the value of m.

   iv). If $(m+1)x^2+2(m+3)x+(m+8) = 0$ has equal roots, then find the value of m
5. i). Show that the equation $2x^2-6x+7 = 0$ has no real root

   ii). Show that the roots of $(x-a)(x-b) = h^2$ are always real

   iii). If p, q, r are rational then show that the roots of $x^2-2px+p^2-q^2+2qr-r^2 = 0$ are rational
6. i). If the quadratic equations $ax^2+2bx+c = 0$ and $ax^2+2cx+b = 0$ $(b\neq c)$ have a common root then show that $a+4b+4c = 0$.

   ii). If $x^2+bx+c = 0$ and $x^2+cx+b = 0$ $(b\neq c)$ have a common root, then show that $b+c+1 = 0$

   iii). If $x^2-6x+5 = 0$, $x^2-12x+p = 0$ have a common root find p

   iv). If $x^2-6x+5 = 0$, $x^2-3ax+35 = 0$ have a common root then find a
7. Find the quadratic equation for which

   i). the sum of the roots is 1 and the sum of squares of the roots is 13

   ii). the sum of the roots is 7 and the sum of squares of the roots is 25
8. State the sign of the following expressions for x is real

   i). x^2-5x+4 ii). x^2-5x+6 iii). x^2+x+1 iv). x^2-x+3
9. For what values of x the following expression is positive

   i). x^2-5x+6 ii). $x^2-5x-14$ iii). $x^2-5x+14$ iv). $3x^2+4x+4$
10. For what values of x the following expression is negative

    i). x^2-5x-6 ii). $2x^2+5x-3$ iii). $x^2-7x+10$ iv). $15+4x-3x^2$

    v). $2x-3-6x^2$
11. Find the maximum or minimum of the following expressions

    i). $2x-7-5x^2$ ii). $x^2 - x +7$ iii). $2x+5- 3x^2$
12. Find the set of solutions of $x^2+x - 12 \leq 0$ by algebraic method

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QNO : 5 IN IPE (AP/TS)

4. THEORY OF EQUATIONS

1. i). If 1, 1, α are the roots of $x^3-6x^2+9x-4=0$ then find α

   ii). If the product of the roots of $4x^3+16x^2-9x-a=0$ is 9 then find a

   iii). If -1, 2, α are the roots of $2x^3+x^2-7x-6=0$ then find α

   iv). If 1, -2, 3 are the roots of $x^3-2x^2+ax+6=0$ then find a

   v). If α,β, 1 are the roots of $x^3-2x^2-5x+6=0$ then find α,β

   vi). If α,β,γ are the roots of $4x^3-6x^2+7x+3=0$ then find $\alpha\beta+\beta\gamma+\gamma\alpha$

2. Form the (monic) polynomial equation of degree 3 whose roots are

   i). 2, 3, 6 ii). 1,-1, 3

3. Form the polynomial equation with rational coefficients and whose roots are

   i). 0,1, -3/2, -5/2 ii). $1\pm\sqrt{3}$, 2, 5 iii). $1\pm 2i$, 4, 2 iv). $1\pm i$, $-1\pm i$

   v). $2\pm\sqrt{3}$, $1\pm 2i$ vi). $2\pm 3i$, $1\pm i$ vii). $4\pm\sqrt{3}$, $2\pm i$ viii). 3, 2, $1\pm i$

   ix). $1\pm i$, $1\pm i$ x). 0,0,2,2,-2,-2

4. If 1, 2, 3, and 4 are the roots of $x^4+ax^3+bx^2+cx+d=0$ then find the values of a, b, c, and d

5. If α,β and γ are the roots of $x^3+px^2+qx+r=0$ then find

   i). $\alpha^2+\beta^2+\gamma^2$ ii). $\alpha^3+\beta^3+\gamma^3$ iii). $\frac{1}{\alpha}+\frac{1}{\beta}+\frac{1}{\gamma}$ iv). $\beta^2\gamma^2+\alpha^2\beta^2+\gamma^2\alpha^2$

6. Find the i). sum of the squares,

   ii). sum of the cubes, of the roots of $x^3-px^2+qx-r=0$ in terms of p, q, r

7. If α,β,γ are the roots of $x^3-2x^2+3x-4=0$ then find $\alpha^2\beta^2+\beta^2\gamma^2+\gamma^2\alpha^2$

8. Show that the condition that the roots of $x^3+3px^2+3qx+r=0$ may be in

   i). A.P is $2p^3-3pq+r=0$ ii). G.P is $p^3r=q^3$

9. Given that '2' is a root of $x^3-6x^2+3x+10=0$ find the other roots.

10. Find the roots of $x^3-3x^2-16x+48=0$ by trial and error method

11. Find the equation whose roots are the negatives of the roots of

    i). $x^4+5x^3+11x+3=0$ ii). $x^4-6x^3+7x^2-2x+1=0$

    iii). $x^7+3x^5+x^3-x^2+7x+2=0$.

12. Find the equation whoose roots are

    i). 2 times the roots of

    ii). 3 times the roots of

    iii). 3 times the roots of $6x^4-7x^3+8x^2-7x+2=0$

13. Find the equation whose roots are the reciprocals of the roots of

    i). $x^4+3x^3-6x^2+2x-4=0$ ii). $x^5+11x^4+x^3+4x^2-13x+6=0$

    iii). $x^4-3x^3+7x^2+5x-2=0$

14. Find the equation whose roots are the squares of the roots of

    i). $x^3+3x^2-7x+6=0$ ii). $x^3-x^2+8x-6=0$

15. Find the equation whose roots are the cubes of the roots of $x^3+3x^2+2=0$

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16. From the equation whose roots are 'm' times the roots of the equation $x^3+\frac{1}{4}x^2-\frac{1}{16}x+\frac{1}{72}=0$

17. Show that $2x^3+5x^2+5x+2=0$ is a reciprocal equation of class one.

QNO : 6 IN IPE (AP/TS)
5. PERMUTATIONS

1. Prove that ${}^{n}P_{r}=n.{}^{(n-1)}P_{(r-1)}$
2. i). If ${}^{n}P_{3}=1320$, find n
   ii). If ${}^{n}P_{4}=1680$, find n
   iii). If ${}^{12}P_{r}=1320$, find r
   iv). If ${}^{12}P_{5}+5\ {}^{12}P_{4}={}^{13}P_{r}$, find r
   v). If ${}^{n}P_{7}=42\ {}^{n}P_{5}$ find n
   vi). If ${}^{(n+1)}P_{5}:{}^{n}P_{5}=3:2$, find n
   vii). If ${}^{(n+1)}P_{5}:{}^{n}P_{6}=2:7$, then find n
3. Find the number of ways of arranging the letters of the word 'TRIANGLE' so that the relative positions of the vowels and consonants are not disturbed.
4. Find the number of ways of arranging the letters of the word 'MONDAY' so that no vowel occupies even place.
5. Find the number of ways of arranging 5 boys and 4 girls in a row so that the row
   i). begins and ends with boys
   ii). begins with a boy and ends with a girl.
6. Find the number of 5 letter words that can be formed using the letters of the word 'RHYME' if each letter can be used any number of times.
7. Find the number of 5 letter words that can be formed using the letters of the word 'NATURE' that begin with N when repetition is allowed.
8. Find the number of 5 letter words that can be formed using the letters of the word MIXTURE which begin with an vowel when repetitions are allowed. V
9. Find the number of 5 letter words that can be formed using the letters of the word 'EXPLAIN' that begin and end with an vowel when repetitions are allowed.
10. Find the number of 4 letter words using the letters of the word PISTON in which at least one letter is repeated.
11. Find the number of
    i). 6
    ii). 7 letter palindromes that can be formed using the letters of the word EQUATION
12. Find the number of ways in which 4 letters can be put in 4 addressed envolepes so that no letter goes into the envelope meant for it.
13. Find the number of injections of a set A with 5 elements into a set B with 7 elements.
14. Find the number of functions from a set A with 5 elements into a set B with 4 elements.
15 Find the number of ways of arranging the letters of the word
   i). INTERMEDIATE
   ii). MATHEMATICS
   iii). INDEPENDENCE.
16. i). Find the number of chains that can be prepared using 7 different coloured beads
    ii). Find the number of ways of preparing a chain with 6 different coloured beads

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17. Find the number of ways of arranging 8 persons around a circular table if two particular persons wish to sit together.

18. Find the number of ways of arranging 7 guests and a host around a circle if 2 particular guests wish to sit on either side of the host.

19. Find the number of ways of arranging 4 boys and 3 girls around a circle so that all the girls sit together.

20. Find the number of ways of arranging 7 gents and 4 ladies around a circular table if no two ladies wish to sit together

21. Find the number of different ways of preparing a garland using 7 distinct red roses and 4 distinct yellow roses such that no two yellow roses come together.

22. Find the number of ways of preparing a garland with 3 yellow, 4 pink, and 2 red roses of different sizes such that the two red roses come together.

23. A chain of beads is to be prepared using 6 different red coloured beads and 3 different blue coloured beads. In how many ways can this be done if no two blue coloured beads come together.

QNO : 7 IN IPE (AP/TS)
5. COMBINATIONS

1. Prove that ${}^nC_r + {}^nC_{r-1} = {}^{(n+1)}C_r$

2. i). If ${}^nC_4 = 210$, find n   ii). If $10\ {}^nC_2 = 3\ {}^{n+1}C_3$, find n

3. i).If ${}^{15}C_{2r-1} = {}^{15}C_{2r+4}$, find r   ii).If ${}^{12}C_{r+1} = {}^{12}C_{3r-5}$, find r
   iii). If ${}^{12}C_{s+1} = {}^{12}C_{2s-5}$, find s   iv). If ${}^{17}C_{2t+1} = {}^{17}C_{3t-5}$, find t

4. i). If ${}^nC_{21} = {}^nC_{27}$, then find ${}^{50}C_n$ ii). If ${}^nC_4 = {}^nC_6$, then find n
   iii). If ${}^nC_5 = {}^nC_6$, then find ${}^{13}C_n$

5. Find the value of ${}^{10}C_5 + 2.\ {}^{10}C_4 + {}^{10}C_3$ 6. If ${}^9C_3 + {}^9C_5 = {}^{10}C_r$, then find r

7. If ${}^nP_r = 5040$ and ${}^nC_r = 210$, find n and r.

8. i). Find the number of ways of selecting 4 boys and 3 girls from a group of 8 boys and 5 girls
   ii). Find the number of ways of selecting 3 girls and 3 boys out of 7 girls and 6 boys.

9. Find the number of ways of selecting 3 vowels and 2 consonants from the letters of the word EQUATION.

10. 14 persons are seated at a round table. Find the number of ways of selecting two persons who are not seated adjacent to each other.

11. To pass an examination a student has to pass in each of the three papers. In how many ways can a student fail in the examination.

12. Find the number of positive integral divisors of 1080

13. Find the number of diagonals of a polygon with 12 sides

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QNO : 8 IN IPE (AP/TS)

6. BINOMIAL THEOREM

1. Find i). 6th term of $\left(\frac{2x}{3}+\frac{3y}{2}\right)^9$ ii). 7th term of $\left(\frac{4}{x^3}+\frac{x^2}{2}\right)^{14}$
2. Find the coefficient of

   i). x^{-7} in $\left(\frac{2x^2}{3}-\frac{5}{4x^5}\right)^7$ ii). x^{11} in $\left(2x^2+\frac{3}{x^3}\right)^{13}$
3. If the coefficient of (2r+4)th and (3r+4)th terms in the expansion of $(1+x)^{21}$ are equal, find r
4. If A and B are the coefficients of x^n in the exapansion of $(1+x)^{2n}$ and $(1+x)^{2n-1}$ respectively then find the value of A/B
5. Find the number of terms in the expansion of $(2x+3y+z)^7$
6. Find the middle term in the expansion of $\left(\frac{3x}{7}-2y\right)^{10}$
7. If $^{22}C_r$ is the largest binomial coefficient in the expansion of $(1+x)^{22}$ find the value of $^{13}C_r$
8. i). $C_0 + 2\,C_1 + 2^2\,C_2 + ... + 2^n\,C_n = 3^n$

   ii). $C_0 + 3\,C_1 + 3^2\,C_2 + ... + 3^n\,C_n = 4^n$
9. If $(1+x+x^2)^n = \alpha_0+\alpha_1 x+\alpha_2 x^2+ .... +\alpha_{2n} x^{2n}$ then prove that $\alpha_0+\alpha_1+\alpha_2+.......+\alpha_{2n}=3^n$
10. Find the set of x for which the binomial expansions for the following are valid.

    i). $(3-4x)^{3/4}$ ii). $(2+5x)^{-1/2}$ iii). $(7+3x)^{-5}$ iv). $(4 - x/3)^{-1/2}$

    v). $(5+x)^{3/2}$ vi). $(2+3x)^{-2/3}$
11. Find the 8th term of $\left(1-\frac{5x}{2}\right)^{-3/5}$

QNO : 9 IN IPE (AP/TS)

8. MEASURES OF DISPERSION

1. Define Range and find the range for an ungrouped data 38, 70, 48, 40, 42, 55, 63, 46, 54, 44
2. Find the mean deviation about the mean for the following data

   i). 6, 7, 10, 12, 13, 4, 12, 16. ii). 3, 6, 10, 4, 9, 10

   iii). 38, 70, 48, 40, 42, 55, 63, 46, 54, 44.
3. Find the mean deviation about median for the data

   i). 6, 7, 10, 12, 13, 4, 12, 16 ii). 4, 6, 9, 3, 10, 13, 2

   iii). 13, 17, 16, 11, 13, 10, 16, 11, 18, 12, 17
4. Find the mean, variance and standard deviation of the data

   i). 5, 12, 3, 18, 6, 8, 2, 10 ii). 6, 7, 10, 12, 13, 4, 8, 12
5. The variance of 20 observations is 5. If each observation is multiplied by 2, find the new variance of the resulting observations.

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QNO : 10 IN IPE (AP/TS)

10. RANDOM VARIABLES & ROBABILITY DISTRIBUTION

1. A random variable X has the range {1,2,3 ..... }. If $P(X=k)=\frac{c^k}{k!}$ for k = 1,2 ..... then find 'c' and P(0<X<3)
2. A random variable X has its range {0, 1, 2, 3.....}. If $P(X=k)=\frac{c(k+1)}{2^k}$ for $k=0,1,2$ ..... then find c
3. find the constant c so that $P(x) = c(2/3)^x$, x = 1, 2, 3, ... is the probability distribution function of a discrete random variable X
4. Let X be a random variable such that P(X = -2) = P(X = -1) = P(X = 2) = P(X = 1) = 1/6 and P(X=0) = 1/3 then find the mean of X
5. The mean and variance of a binomial distribution are 4 and 3 respectively.Find the probability of success p, the number of trials n and $P(X \geq 1)$
6. The mean and variance of a binomial variate are 2.4, 1.44 respectively.Find the parameters,
7. For a binomial distribution with mean 6 and variance 2, find the first two terms of the distribution
8. If the difference between the mean and variance of a binamial variate is 5/9 then find the probability for the event of 2 success when the experiment is conducted 5 times
9. The probability that a person chosen at random is left handed (in hand writing) is 0.1.What is the probability that in a group of ten people there is one, and only one, who is left handed
10. On an average rain falls on 12 days in every 30 days, find the probability that, rain will fall on just 3 days of a given week
11. A poisson variable satisfies P(X = 1) = P(X = 2). Find P(X = 5)

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QNO : 11 IN IPE (AP/TS)

1. COMPLEX NUMBERS

1. i). If $z = 2 - i\sqrt{7}$, then show that $3z^3 - 4z^2 + z + 88 = 0$

   ii). If $z = 3-5i$, show that $z^3-10z^2+58z-136 = 0$

2. i). If $x + iy = \dfrac{1}{1+\cos\theta + i\sin\theta}$, show that $4x^2 - 1 = 0$

   ii). If $x + iy = \dfrac{3}{2+\cos\theta + i\sin\theta}$, then show that $x^2+y^2 = 4x-3$

   iii). If $u+iv = \dfrac{2+i}{z+3}$, where $z = x+iy$, find the values of u and v

   iv). if $(x - iy)^{1/3} = a - ib$, show that $\dfrac{x}{a}+\dfrac{y}{b} = 4(a^2 - b^2)$

3. Prove that i). $\dfrac{2-i}{(1-2i)^2}$ and $\dfrac{-(2+11i)}{25}$ are conjugate complex numbers.

   ii). $z_1 = \dfrac{2+11i}{25}$, $z_2 = \dfrac{-2+i}{(1-2i)^2}$ are conjugate complex numbers

4. Find the locus of $z = x+iy$, if i). $|z-3+i| = 4$ ii). $|z-2-3i| = 5$

5. Find the locus of $z = x + iy$ if

   i). The real part of $\dfrac{z+1}{z+i}$ is 1 ii). $\dfrac{z-i}{z-1}$ is purely imaginary

   iii). Re $\dfrac{z-4}{z-2i} = 0$ iv). Amplitude of $\dfrac{z-2}{z-6i}$ is $\dfrac{\pi}{2}$

6. A variable complex number $z = x + iy$ is such that the amplitude of $\dfrac{z-1}{z+1}$ is always equal to $\dfrac{\pi}{4}$

   Show that $x^2 + y^2-2y -1 = 0$

7. Show that in the Argand plane, the points represented by the complex numbers

   i). $2+2i$, $-2-2i$, $-2\sqrt{3}+2\sqrt{3}\,i$ from an equilateral triangle

   ii). $-2+7i$, $-\dfrac{3}{2}+\dfrac{1}{2}i$, $4 - 3i$, $\dfrac{7}{2}(1+i)$ taken in order from a rhombus

   iii). $2+i$, $4+3i$, $2+5i$, $3i$ taken in order from a square

QNO : 12 IN IPE (AP/TS)

3. QUADRATIC EXPRESSIONS

1. If x_1, x_2 are the roots of $ax^2+bx+c = 0$ then find the value of $(ax_1+b)^{-2}+(ax_2+b)^{-2}$

2. If $c^2 \neq ab$ and the roots of $(c^2-ab)x^2-2(a^2-bc)x+(b^2-ac) = 0$ are equal then.show that $a = 0$ (or) $a^3+b^3+c^3 = 3abc$

3. If $x^2+4ax+3 = 0$ and $2x^2+3ax-9 = 0$ have a common root, then find the value of ‘a’ and the common root

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4. Solve $4^{x-1} - 3\,2^{x-1} + 2 = 0$
5. Solve the equation $2x^4+x^3-11x^2+x+2 = 0$
6. Find the range of the following expressions if x is real

   i). $\dfrac{(x-1)(x+2)}{x+3}$ ii). $\dfrac{2x^2-6x+5}{x^2-3x+2}$ iii). $\dfrac{x+2}{2x^2+3x+6}$ iv). $\dfrac{x^2+x+1}{x^2-x+1}$
7. If x is real , prove that

   i). $\dfrac{x}{x^2-5x+9}$ lies between $-\dfrac{1}{11}$ and 1

   ii). $\dfrac{x^2+34x-71}{x^2+2x-7}$ does't lie between 5 and 9

   iii). $\dfrac{1}{3x+1}+\dfrac{1}{x+1}-\dfrac{1}{(3x+1)(x+1)}$ does not lie between 1 and 4
8. If x is real,find the maximum value of $\dfrac{x^2+14x+9}{x^2+2x+3}$
9. If $\dfrac{x-p}{x^2-3x+2}$ takes all real values for $x \in R$ show that $1<p<2$

QNO : 13 IN IPE (AP/TS)
5. PERMUTATIONS

1. i). if ${}^{18}P_{r-1} : {}^{17}P_{r-1} = 9 : 7$, find r. ii). if ${}^{56}P_{r+6} : {}^{54}P_{r+3} = 30800 : 1$ find r
2. Find the number of 4 letter words that can be formed using the letters of the word 'MIRACLE'. How many of them i). begin with an vowel ii). begin and end with vowels iii). end with a consonants
3. Find the number of ways of arranging 5 different Mathematics Books, 4 different Physics Books, and 3 different Chemistry Books such that the books of the same subject are together.
4. Find the number of numbers that are greater than 4000 which can be formed using the digits 0, 2, 4, 6, 8 without repetition.
5. Find the sum of all 4 digited numbers that can be formed using the digits
   i). 1, 3, 5, 7, 9 ii). 1, 2, 4, 5, 6 iii). 0, 2, 4, 7, 8.
6. If the letters of the word PRISON are permuted in all possible ways and the words thus formed are arranged in dictionary order, find the rank of the word 'PRISON'.
7. i). If the letters of the word MASTER are permuted in all possible ways and the words thus formed are arranged in dictinary order, find the rank of the word
   i). REMAST ii). MASTER
8. Find the number of ways of arranging the letters of the word 'SINGING' so that
   i). They bigin and end with I ii). The two G' s come together.
9. If the letter of the word 'EAMCET' are permuted in all possible ways and if the words thus formed are arranged in the dictionary order.Find the rank of the word 'EAMCET'.

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10. A round table conference is attended by 3 Indians, 3 Chinese, 3 Canadians and 2 Americans. Find the number of ways of arranging them at the round table so that the delegates belonging to same coutry sit together.

11. Find the number of ways of arranging 6 red roses and 3 yellow roses of different sizes into a garland. In how many of them i) all the yellow roses are together
ii) no two yellow roses are together

QNO : 14 IN IPE (AP/TS)

5. COMBINATIONS

1. Show that ${}^{(n-3)}C_r + 3\,{}^{(n-3)}C_{(r-1)} + 3\,{}^{(n-3)}C_{(r-2)} + {}^{(n-3)}C_{(r-3)} = {}^{n}C_r$

2. Prove that i). ${}^{25}C_4 + \sum_{r=0}^{4} {}^{29-r}C_3 = {}^{30}C_4$ ii). ${}^{34}C_5 + \sum_{r=0}^{4} {}^{38-r}C_4 = {}^{39}C_5$

3. Show that $\dfrac{{}^{4n}C_{2n}}{{}^{2n}C_n} = \dfrac{[1.3.5.....(4n-1)]}{[1.3.5.....(2n-1)]^2}$

4. Find the number of ways of selecting a cricket team of 11 players from 7 batsmen and 6 bowlers such that there will be at least 5 bowlers in the team.

5. Find the number of ways of forming a committee of 4 members out of 6 boys and 4 girls such that there is at least one girl in the committee.

6. Find the number of ways of forming a committee of 5 members out of 6 Indians and 5 Americans so that always the Indians will be in majority in the committtee.

7. Find the number of ways of forming a committee of 5 members from 6 men and 3 ladies. How many committees contain at least two ladies.

8. Find the number of ways of selecting 11 member cricket team from 7 batsmen, 6 bowlers and 2 wicket keepers with at least 4 bowlers and 2 wicket keepers.

9. A candidate is required to answer 6 out of 10 questions which are devided into two groups A and B each containing 5 questions. He is not permitted to attempt more than four questions from either group. Find the number of different ways in which the candidate can chose 6 questions.

QNO : 15 IN IPE (AP/TS)

7. PARTIAL FRACTIONS

Resolve into partial fractions

1. i). $\dfrac{3x+7}{x^2-3x+2}$ ii). $\dfrac{5x+1}{(x+2)(x-1)}$ iii). $\dfrac{x+4}{(x^2-4)(x+1)}$

2. i). $\dfrac{2x+3}{(x-1)^3}$ ii). $\dfrac{x^2+5x+7}{(x-3)^3}$ iii). $\dfrac{3x^3-8x^2+10}{(x-1)^4}$

3. i). $\dfrac{x^2+13x+15}{(2x+3)(x+3)^2}$ ii). $\dfrac{x^2-x+1}{(x+1)(x-1)^2}$ iii). $\dfrac{1}{(x-1)^2(x-2)}$

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iv). $\frac{3x-18}{x^3(x+3)}$ v). $\frac{2x^2+2x+1}{x^3+x^2}$

4. i). $\frac{x^3}{(x-1)(x+2)}$ ii). $\frac{x^4}{(x-1)(x-2)}$ iii). $\frac{x^3}{(x-a)(x-b)(x-c)}$

iv). $\frac{x^3}{(2x-1)(x+2)(x-3)}$

5. i). $\frac{2x^2+3x+4}{(x-1)(x^2+2)}$ ii). $\frac{x^2-3}{(x+2)(x^2+1)}$ iii). $\frac{3x-1}{(1-x+x^2)(2+x)}$

iv). $\frac{2x^2+1}{x^3-1}$

6. Resolve $\frac{3x^3-2x^2-1}{x^4+x^2+1}$ into partial fractions

7. Find the coefficient of

i). x^3 in the expansion of $\frac{5x+6}{(x+2)(1-x)}$ ii). x^4 in the expansion of $\frac{3x}{(x-2)(x+1)}$

iii). x^n in the expansion of $\frac{x}{(x-1)^2(x-2)}$ iv). x^4 in the expansion of $\frac{3x^2+2x}{(x^2+2)(x-3)}$

v). x^n in the expansion of $\frac{x-4}{x^2-5x+6}$

QNO : 16 IN IPE (AP/TS)
9. PROBABILITY

1. Find the probability that a non - leap year contains i). 53 Sundays
   ii). 52 Sundays only
2. If a number x is selected from natural numbers 1 to 100, find the probability for $x+\frac{100}{x}>29$
3. If two numbers are selected randomly from 20 consecutive natural numbers, find the probability that the sum of the two numbers is i). an even number
   ii). an odd number
4. A bag contains 12 two rupees coins , 7 one rupee coins and 4 half a rupee coins. If three coins are selected at random then find the probability that
   i) The sum of the three coins is maximum
   ii) The sum of three coins is minimum. iii) each coin is of different value.
5. In a box containing 15 bulbs, 5 are defective. If 5 bulbs are selected at random from the box, find the probability of the event, that
   i) none of them is defective . ii) only one of them is defective.
   iii) atleast one of them is defective.
6. State and prove ADDITION THEOREM on probability
   If A, B are two events in a sample space S, then $P(A\cup B) = P(A)+P(B) - P(A\cap B)$

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7. if A,B,C are three events in a sample space S, then
$P(A\cup B\cup C) = P(A) + P(B) + P(C) - P(A\cap B) - P(B\cap C) - P(C\cap A) + P(A\cap AB\cap C)$

8. Find $P(\overline{A})+P(\overline{B})$ if $P(A\cup B) = 0.65$, $P(A\cap B) = 0.15$

9. Suppose A and B are events with $P(A) = 0.5$, $P(B) = 0.4$ and $P(A\cap B) = 0.3$.Find the probability that i). A does not occur,
ii). neither A nor B occurs.

10. For any two events A and B, show that $P(\overline{A}\cap\overline{B}) = 1 - P(A) - P(B) + P(A\cap B)$

11. The probability for a contractor to get a road contract is 2/3 and to get a building contract is 5/9. The probability to get atleast one contract is 4/5. Find the probability that he gets both the contracts.

12. Find the probability of drawing an ace or a spade from a well shuffled pack of 52 cards.

13. In a class of 60 boys and 20 girls, half of the boys and half of the girls know cricket.Find the probability of the event that a person selected from the class is either 'a boy' or 'a girl who knowing cricket'.

14. A,B,C are three horses in a race. The probability of A to win the race is twice that of B, and the probability of B is twice that of C. What are the probabilities of A, B and C to win the race.

15. In a committee of 25 members, each member is proficient either in mathematics, or in statistics or in both. If 19 of these are proficient in mathematics, 16 in statistics, find the probability that a person selected from the committee is proficient in both.

16. i). If one ticket is randomly selected from tickets numbered 1 to 30, then find the probability that the number on the ticket is multiple of 5 or 7.

ii). If one ticket is randomly selected from tickets numbered 1 to 30, then find the probability that the number on the ticket is multiple of 3 or 5.

17. If $P(A) = 0.3$, $P(B) = 0.4$, $P(C) = 0.8$, $P(A\cap B) = 0.08$, $P(A\cap C) = 0.28$, $P(A\cap B\cap C) = 0.09$, $P(A\cup B\cup C)$ is greater than are equal to 0.75, then show that $P(B\cap C)$ lies in [0.23, 0.48]

18. A,B,C are three news papers published from a city. 20% of the population read A, 16% read B, 14% read C, 8% read both A and B, 5% read both A and C, 4% read both B and C and 2% read all the three.Find the percentage of the population who read atleast one news paper.

19. The probabilities of 3 mutually exclusive events are respectively given as
$\frac{1+3p}{3}, \frac{1-p}{4}, \frac{1-2p}{2}$. Prove that $\frac{1}{3}\le p\le\frac{1}{2}$

QNO : 17 IN IPE (AP/TS)
9. PROBABILITY

20. If A, B are two independent events in a sample space, then
i). $\overline{A}, \overline{B}$ are independent. ii). $A\cup B$, C are also independent.

21. If A and B are independent events with $P(A) = 0.2$, $P(B) = 0.5$. Find
i). P(A/B) ii).P(B/A)
iii).$P(A\cap B)$ iv).$P(A\cup B)$

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22. Suppose A and B are independent events with P(A) = 0.6, P(B) = 0.7 Compute
i). $P(A \cap B)$ ii). $P(A \cup B)$
iii). P(B/A) iv). $P(\overline{A} \cap \overline{B})$

23. A fair die is rolled. Consider the events A = {1,3,5}, B = {2,3} and C = {2,3,4,5} find
i). $P(A \cup B)$ ii). P(A/B)
iii). P(A/C) iv). P(B/C)

24. If one card is drawn at random from a pack of cards then show that getting an ace and getting heart are independent events.

25. The probability that a boy A will get a scholarship is 0.9 and that another boy B will get is 0.8 What is the probability that atleast one of them will get the scholorship.

26. A problem in calculus is given to two students A and B whose chances of solving it are 1/3 and 1/4. What is the probability that the problem being solved if both of them try independently.

27. A speaks truth in 75% of the cases and B in 80% of the Cases. What is the probability that their statements about an incident do not match.

28. The probability that Australia wins a match against India in a cricket game is given to be 1/3. If India and Australia play three matches, what is the probability that
i) Australia will loose all the three matches.
ii) Australia will win atleast one match.

29. A,B,C are aiming to shoot a balloon. A will succeed 4 times out of 5 attempts.The chance of B to shoot the balloon is 3 out of 4 and that of C is 2 out of 3.If the three aim the balloon simultaneously.find the probability that atleast 2 of them hit the balloon.

30. In a shooting test the probability of A, B ,C having the targets are 1/2, 2/3, and 3/4 respectively.If all of them fire at the same target, find the probability that
i). only one of them hits the target,
ii). atleast one of them hits the target.

31. A,B are two independent events such that, the probability of both the events to occur is 1/6 and the probability of both the events do not occur is 1/3. Find P(A).

32. If A, B, C are three independent events such that $P(A \cap \overline{B} \cap \overline{C}) = 1/4$, $P(\overline{A} \cap B \cap \overline{C}) = 1/8$, $P(\overline{A} \cap \overline{B} \cap \overline{C}) = 1/4$, then find P(A), P(B), P(C).

33. Let A, B be two events in a sample space S such that $P(A) \neq 0$, $P(B) \neq 0$ then
i). $P(A \cap B) = P(A)\, P(B/A)$
ii). $P(A \cap B) = P(B)\, P(A/B)$

34. Three screws are drawn at random from a lot of 50 screws, 5 of which are defective. Find the probability of the event that all 3 screws are non defective, assuming that the drawing is
i). with replacement b). without replacement.

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QNO : 18 IN IPE (AP/TS)

2. DE MOIVRES THEOREM

1. If n is a positive integer, prove that

i). $(1+i)^n + (1-i)^n = 2^{\frac{n}{2}+1}\cos\frac{n\pi}{4}$ ii). $(1+i)^{2n} + (1-i)^{2n} = 2^{n+1}\cos\frac{n\pi}{2}$

iii). $(p+iq)^{1/n}+(p-iq)^{1/n} = 2(p^2+q^2)^{1/2n}\cos\left(\frac{1}{n}\mathrm{Tan}^{-1}\frac{q}{p}\right)$

iv). $(1+\cos\theta+i\sin\theta)^n+(1+\cos\theta-i\sin\theta)^n = 2^{n+1}\cos^n\frac{\theta}{2}\cos\frac{n\theta}{2}$

2. If α and β are the roots of $x^2 - 2x + 4 = 0$ then show that $\alpha^n + \beta^n = 2^{n+1}\cos\frac{n\pi}{3}$

3. Prove that one of the value of $\left(\frac{1+\sin(\pi/8)+i\cos(\pi/8)}{1+\sin(\pi/8)-i\cos(\pi/8)}\right)^{8/3}$ is -1

4. If $\cos\alpha+\cos\beta+\cos\gamma = 0 = \sin\alpha+\sin\beta+\sin\gamma$ then show that

i). $\cos3\alpha+\cos3\beta+\cos3\gamma = 3\cos(\alpha+\beta+\gamma)$ ii). $\sin3\alpha+\sin3\beta+\sin3\gamma = 3\sin(\alpha+\beta+\gamma)$

iii). $\cos^2\alpha+\cos^2\beta+\cos^2\gamma = 3/2$ iv). $\sin^2\alpha+\sin^2\beta+\sin^2\gamma = 3/2$

5. If $z^2+z+1 = 0$, where z is a complex number then prove that

$$\left(z+\frac{1}{z}\right)^2+\left(z^2+\frac{1}{z^2}\right)^2+\left(z^3+\frac{1}{z^3}\right)^2+\,.....\,+\left(z^6+\frac{1}{z^6}\right)^2 = 12$$

6. if n is an integer and $z = \cos\theta + i\sin\theta$, show that $\frac{z^{2n}-1}{z^{2n}+1} = i\,\mathrm{Tan}\,n\theta$

7. if $x = \cos\alpha+i\sin\alpha$, $y = \cos\beta+i\sin\beta$, show that

i). $x^m y^n+\frac{1}{x^m y^n} = 2\cos(m\alpha+n\beta)$ ii). $x^m y^n-\frac{1}{x^m y^n} = 2i\sin(m\alpha+n\beta)$

8. Solve $(x - 1)^n = x^n$, where n is a positive integer

9. Solve the equation i). $x^{11} - x^7 + x^4 - 1 = 0$ ii). $x^9 - x^5 + x^4 - 1 = 0$

QNO : 19 IN IPE (AP/TS)

4. THEORY OF EQUATIONS

1. Solve the equation

i). $x^4+2x^3-5x^2+6x+2 = 0$, given that $1+i$ is a root

ii). $x^4-4x^2+8x+35 = 0$, given that $2+i\sqrt{3}$ is a root

iii). $x^4-9x^3+27x^2-29x+6 = 0$, one root being $2-\sqrt{3}$

iv). $x^4+2x^3-16x^2-22x+7 = 0$, one root being $2-\sqrt{3}$

2. Solve the equations if the roots being in A.P

i). $x^3-3x^2-6x+8 = 0$ ii). $4x^3-24x^2+23x+18 = 0$

iii). $8x^3-36x^2-18x+81 = 0$

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3. Solve the equations if the roots being in G.P

i). $x^3-7x^2+14x-8 = 0$ ii). $3x^3-26x^2+52x-24 = 0$

4. Solve the equation

i). $2x^3+3x^2-8x+3 = 0$ one root being double(twice) the another root

ii). $18x^3+81x^2+121x+60= 0$ one root being half the sum of the other two

5. Solve the equation

i). $x^4+x^3-16x^2-4x+48 = 0$ the product of two of the roots being 6

ii). $x^4+4x^3-2x^2-12x+9 = 0$ If it has two pairs of equal roots

6. Find the roots of $x^4 - 16x^3 + 86x^2 - 176x + 105 = 0$

7. Show that $x^5-5x^3+5x^2-1 = 0$ has three equal roots and find this root

8. Solve the equation

i). $x^4 -10x^3+26x^2-10x+1 = 0$ ii). $6x^4 - 35x^3+62x^2 - 35x+6 = 0$

iii). $2x^5+x^4-12x^3-12x^2+x+2 = 0.$ iv). $6x^6-25x^5+31x^4-31x^2+25x-6 = 0.$

v). $x^5-5x^4+9x^3-9x^2+5x-1 = 0$

9. Find the equation whose roots are the translates of the roots of the equation

i). $x^4-5x^3+7x^2-17x+11 = 0$ by -2 ii). $x^5+4x^3-x^2+11 = 0$ by -3

QNO : 20 IN IPE (AP/TS)
6. BINOMIAL THEOREM

1. i). Find the sum of the coefficients of x^{32} and x^{-18} in the expansion of $\left(2x^3 - \frac{3}{x^2}\right)^{14}$

ii). If the coefficient of x^{10} in $(ax^2+ 1/bx)^{11}$ is equal to the coefficient of x^{-10} in $\left(ax - \frac{1}{bx^2}\right)^{11}$

Find the relation between a and b where a and b are real numbers.

2. i). Find the values of a, x, n if the 2^{nd}, 3^{rd}, 4^{th} terms in $(a+x)^n$ are 240,720,1080

ii). If 36, 84, 126 are three successive bionamial coefficients in the expansion of $(1+x)^n$ then find n

3. i). If the coefficients of rth, (r+1)th and (r+2)nd terms in the expansion of $(1+x)^n$ are in A.P then show that $n^2 - (4r +1) n + 4r^2 - 2 = 0$

ii). If the coefficients of x^9, x^{10}, x^{11} in the expansion of $(1+x)^n$ are in A.P. then prove that $n^2 - 41n + 398 = 0$

4. If a_1, a_2, a_3, a_4 are the coefficients of four consecutive terms respectively in the expansion of $(1+x)^n$ where n is a positive integer then show that $\frac{a_1}{a_1+a_2} + \frac{a_3}{a_3+a_4} = \frac{2a_2}{a_2+a_3}$

5. If the sum of odd terms and the sum of even terms in the expansion of $(x+a)^n$ are p and q respectively then show that

i). $p^2-q^2 = (x^2-a^2)^n$ ii). $4pq = (x+a)^{2n} - (x-a)^{2n}$

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6. If $(7+4\sqrt{3})^n = I+F$ where I and n are positive integers and 0<F<1 then show that I is an odd integer and (I+F)(1-F) = 1

7. If C_r denotes ${}^nC_r$, then $C_0+\frac{C_1}{2}x+\frac{C_2}{3}x^2+......+\frac{C_n}{n+1}.x^n = \frac{(1+x)^{n+1}-1}{(n+1)x}$ and hence prove that

i). $C_0+\frac{C_1}{2}+\frac{C_2}{3}+......+\frac{C_n}{n+1} = \frac{2^{n+1}-1}{n+1}$.

ii). $C_0+\frac{3}{2}C_1+\frac{9}{3}C_2+\frac{27}{4}C_3+........+\frac{3^n}{n+1}C_n = \frac{4^{n+1}-1}{3(n+1)}$

8. For r = 0, 1, 2, ... n, prove that $C_0C_r+C_1C_{r+1}+C_2C_{r+2}+.....+C_{n-r}C_n = {}^{2n}C_{n+r}$ and hence prove that i). $C_0^2+C_1^2+C_2^2+......+C_n^2 = {}^{2n}C_n$

ii). $C_0C_1+C_1C_2+C_2C_3+......+C_{n-1}C_n = {}^{2n}C_{n+1}$

9. Show that $\sum_{r=1}^{n} r^3\left(\frac{{}^nC_r}{{}^nC_{r-1}}\right)^2 = \frac{n(n+1)^2(n+2)}{12}$

QNO : 21 IN IPE (AP/TS)

6. BINOMIAL THEOREM

1. Show that

i). $1+\frac{1}{3}+\frac{1.3}{3.6}+\frac{1.3.5}{3.6.9}$ .... $\infty = \sqrt{3}$

ii). $1-\frac{4}{5}+\frac{4.7}{5.10}-\frac{4.7.9}{5.10.15}+...... = \frac{\sqrt[3]{625}}{16}$

iii). $\frac{7}{5}\left(1+\frac{1}{10^2}+\frac{1.3}{1.2}.\frac{1}{10^4}+\frac{1.3.5}{1.2.3}.\frac{1}{10^6}+....\right) = \sqrt{2}$

2. Show that

i). $\frac{3}{4}+\frac{3.5}{4.8}+\frac{3.5.7}{4.8.12}+........\infty = \sqrt{8}-1$

ii). $\frac{3.5}{5.10}+\frac{3.5.7}{5.10.15}+\frac{3.5.7.9}{5.10.15.20}+......\infty = \frac{5\sqrt{5}}{3\sqrt{3}}-\frac{8}{5}$

iii). $\frac{3}{4.8}-\frac{3.5}{4.8.12}+\frac{3.5.7}{4.8.12.16}-.....\infty = \sqrt{\frac{2}{3}}-\frac{3}{4}$

3. i). If $x = \frac{1}{5}+\frac{1.3}{5.10}+\frac{1.3.5}{5.10.15}+......\infty$, then show that $3x^2+6x = 2$

ii). If $x = \frac{5}{2!.3}+\frac{5.7}{3!.3^2}+\frac{5.7.9}{4!.3^3}+....\infty$, then show that $x^2+4x = 23$

iii). If $x = \frac{1.3}{3.6}+\frac{1.3.5}{3.6.9}+\frac{1.3.5.7}{3.6.9.12}+.....\infty$, then show that $9x^2+24x = 11$

4. If $t = \frac{4}{5}+\frac{4.6}{5.10}+\frac{4.6.8}{5.10.15}+.......\infty$, then prove that $9t = 16$

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8. MEASURES OF DISPERSION

1. Find the mean deviation about the mean for the data

x_i :	2	5	7	8	10	35
f_i :	6	8	10	6	8	2

2. Find the mean deviation about the median for the data

x_i :	3	6	9	12	13	15	21	22
f_i :	3	4	5	2	4	5	4	3

3. i). Find the mean deviation about the mean for the data

Marks obtained :	0-10	10-20	20-30	30-40	40-50
Number of Students :	5	8	15	16	6

ii). Find the mean deviation about the mean for the data

Sales (In Rs. thousand) :	40-50	50- 60	60-70	70-80	80-90	90-100
Number of Companies :	5	15	25	30	20	5

4. Find the mean deviation about mean for the data by shortcut method

Classes :	0-10	10-20	20-30	30-40	40-50	50-60	60-70
Frequency :	6	5	8	15	7	6	3

5. i). Calculate the mean deviation about median age for the age distribution of 100 persons given below.

Age(years) :	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60
No of Workers :	120	125	175	160	150	140	100	30

ii). Find the mean deviation about median for the data

Marks :	0-10	10-20	20-30	30-40	40-50	50-60
No of Girls :	6	8	14	16	4	2

6. Find the mean and variance of the following data

x_i :	4	8	11	17	20	24	32
f_i :	3	5	9	5	4	3	1

7. Find mean, variance and standard deviation for the following data by **shortcut** method.

Class :	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Frequency :	3	7	12	15	8	3	2

8. The mean of 5 observations is 4.4 and their variance is 8.24.If three of the observations are 1, 2 and 6, find the other two observations.

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9. PROBABILITY

1. State and Prove BAYE'S THEOREM
2. A bag B_1 contains 4 white and 2 black balls. Another bag B_2 contains 3 white and 4 black balls. A bag is drawn at random and a ball is chosen at random from it. Then what is the probability that the ball drawn is white.

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3. There are 3 black and 4 white balls in one bag, 4 black and 3 white balls in the second bag. A die is rolled and the first bag is selected if it is 1 or 3, and the second bag for the rest. Find the probability of drawing a black ball from the selected bag.

4. Suppose that an urn B_1 contains 2 white and 3 black balls, and another urn B_2 contains 3 white and 4 black balls. One urn is selected at random and a ball is drawn from it. If the ball drawn is found black, find the probability that the urn chosen was B_1.

5. Three boxes B_1, B_2, B_3 contains 2 white, 1 black, and 2 red balls; 3 white, 2 black, and 4 red balls; 4 white, 3 black, and 2 red balls respectively. A die is rolled and B_1 is selected if the number is 1 or 2; B_2 if the number is 3 or 4; B_3 if the number is 5 or 6.Having choosen a box in this way, a ball is choosen at random from this box.If the ball drawn is found to be red, find the probability that it is drawn from box B_2.

6. Three urns have the following composition of balls Urn I : 1 White, 2 Black ; Urn II : 2 White, 1 Black; Urn III : 2 White, 2 Black. One of the urn is selected at randomand a ball is drawn . It turns out to be white. Find the probability that it came from urn III.

7. Three boxes numbered I, II, III contain 1 white, 2 black, and 3 red balls; 2 white, 1 black, and 1 red balls; 4 white, 5 black, and 3 red balls respectively. One box is randomly selected and a ball is drawn from it.If the ball is red then find the probability that it is from box II.

8. Two persons A and B are rolling a die on the condition that the person who gets 3 will win the game. If A starts ther game then find the probabilities of A and B respectively to win the game.

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10. RANDOM VARIABLES & ROBABILITY DISTRIBUTION

1. The probability distribution of a random variable X is given below.

$X = x$	1	2	3	4	5
$P(X = x)$	k	2k	3k	4k	5k

Find the value of k, mean & variance of X.

2. A random variable X has the following probability distribution.

$X = x$	-2	-1	0	1	2	3
$P(X = x)$	0.1	k	0.2	2k	0.3	k

Then find K , mean and variance

3. A random variable X has the following probability distribution.

$X = x$	0	1	2	3	4	5	6	7
$P(X = x)$	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

Then find

i). K. ii). The mean and

iii). $P(0<X<5)$

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4. The range of a random variable X is $\{0,1,2\}$ given that $P(X=0)=3c^3$, $P(X=1)=4c-10c^2$, $P(X=2)=5c-1$ where c is a constant. Find
   i) The value of c ii) $P(X<1)$
   iii) $P(1<X\leq 2)$ iv) $P(0<X\leq 3)$

5. Two dice are rolled at random. Find the probability distribution of the sum of the numbers on them. Find the mean of the random variable.

6. A cubical die is thrown.Find the mean and variance of X, given the number on the face that shows up.

7. 8 coins are tossed simultaneously. Find the probability of getting
   i). 6 heads
   ii). atleast 6 heads

8. In the experiment of tossing a coin 'n' times if the variable X denotes the number of heads and $P(X=4)$, $P(X=5)$, $P(X=6)$ are in A.P. then find the value of 'n'.

9. One in 9 ships is likely to be wrecked, when they are set on sail. When 6 ships set on sail find the probability for
   i). atleast 1 will arrive safely
   ii). exactly 3 will arrive safely.

10. Find the probability of guessing at least 6 out of 10 of answers in True or false type examination

May - 2022 AP Question Paper With Key

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